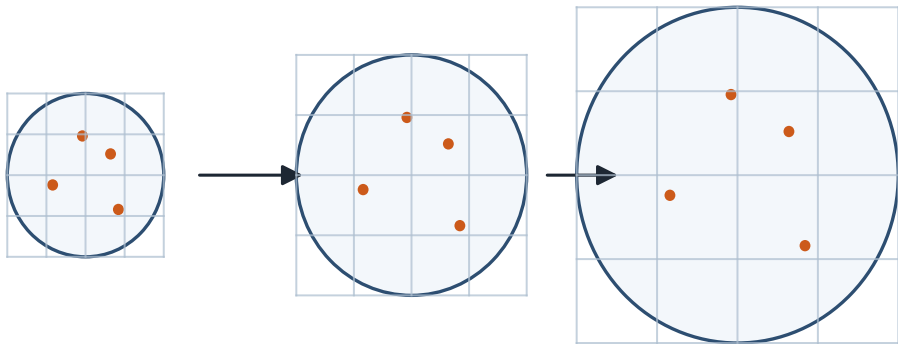


physical separations obey $D_{\text{phys}}(t) = a(t) D_{\text{com}}$



early: $a(t) < 1$

today: $a_0 = 1$

later: $a(t) > 1$

photon wavelengths stretch with the same factor: $1 + z = a_0/a(t_{\text{em}})$